Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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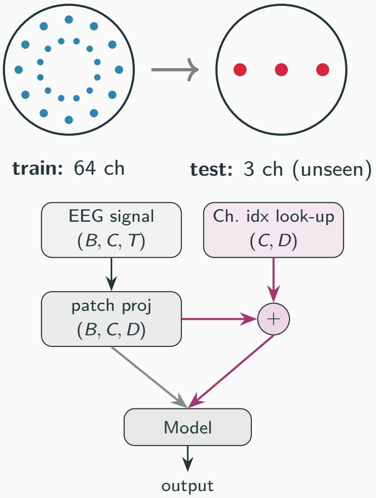
Problem: EEG is non-stationary – and the geometry is what changes

EEG varies on **three geometric axes**:

- **Session** — cap shifts day to day. *(motivates)*
- **Subject** — head sizes differ. *(motivates)*
- **Montage** — 2 to 256 electrodes in different layouts. ← **what I test**

Codex models (LaBraM, EEGPT): identity-indexed embeddings — **cannot even run** on an unseen layout.
Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



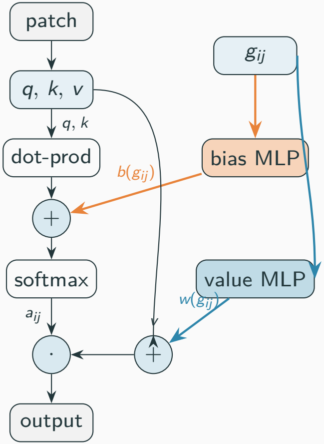
Solution: spatial attention conditioned on g_{ij}

$$g_{ij} = \left[\, p_i, \quad p_j, \quad p_i - p_j, \quad \|p_i - p_j\| \,\right] \in \mathbb{R}^{10}$$

Example: $g_{\texttt{C3}\to\texttt{Cz}}$ (normalized coords)

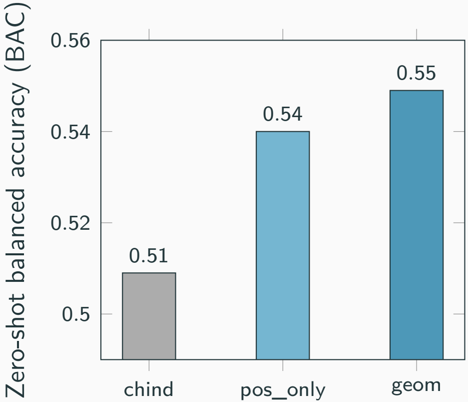
p_i (C3)	(x, y, z)	$(-0.71,\ 0.00,\ 0.59)$
p_j (Cz)	(x, y, z)	$(\ \ 0.00,\ 0.00,\ 0.95)$
$p_i - p_j$	disp.	$(-0.71,\ 0.00,\ -0.36)$
$\|p_i - p_j\|$	dist.	0.80

$3 + 3 + 3 + 1 = 10$ dims. **pos_only** ablation keeps just $[p_i, p_j] \in \mathbb{R}^6$.



bias path only = G1, **value path** only = G2, both = G3.
Ablation: the signal is in the **value path** — G2 ≈ G3 ≫ G1 (G1 ≈ no geometry).

Headline: at scale, geometric beats chind on zero-shot transfer



cells = row-normalized recall (%); diag mean = BAC

geom (G2, full) (BAC 0.549)

	Pred L	Pred R
T.L	56.4	43.6
T.R	46.6	53.4

pos_only (BAC 0.540)

	Pred L	Pred R
T.L	53.8	46.2
T.R	45.8	54.2

78 PhysioNet + 78 Sleep-EDFx pretrain → zero-shot to BCIC-2B

Conclusion: a clean positive, with clear edges

What I found:

- Geometry-conditioned attention **transfers zero-shot** to an unseen montage and **beats chind significantly** once given enough data.
- Robust to descriptor reduction (pos_only also significant).

Honest limits:

- Significant on **one** held-out montage (BCIC); small-corpus tests point the right way but are not significant (9-subject ceiling, underpowered).

Future work: second distinct-montage held-out (e.g. 128 ch); larger corpus; implement the codex baseline.

Did I run what the proposal promised?

Proposed experiment	Outcome
E2(c) — cross-montage zero-shot	✓ **the headline win, significant**
E3 — where g_{ij} enters (G1/G2/G3)	✓ **signal is value-path: $\mathbf{G2 \approx G3 \gg G1}$**
E4 — channel-independent baseline	✓ **promoted to my baseline**
E1 — in-distribution vs. codex	codex can't run on a new montage
E2(a) — cross-session	session drift is physiological, not geometric
E2(b) — cross-subject / head size	no public head-size-varied dataset

Top: delivered. **Bottom:** out of reach for a course project — reported as limits.

`github.com/tianxin-scu/geometric-eeg-ssl`

Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – EEG foundation model; training recipe.
- Jiang et al. **LaBraM**. ICLR 2024 – codex (identity-indexed) baseline.
- Yang et al. **BIOT**. NeurIPS 2023 – channel-independent baseline.
- Schalk et al. 2004 – PhysioNet MI dataset.
- Tangermann et al. 2012 – BCIC-2B dataset.
- Kemp et al. 2000 – Sleep-EDFx dataset.

Full reference list and per-experiment details in the project report.